Vivisection

Albert Leffingwell

Project Gutenberg

Vivisection

Albert Leffingwell

Project Gutenberg

Title: Vivisection

Author: Albert Leffingwell

Release date: April 17, 2010 [eBook #32033]
Most recently updated: January 6, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/32033

Credits: Produced by The Online Distributed Proofreading Team at https://www.pgdp.net (This file was produced from images generously made available by The Internet Archive/American Libraries.)

VIVISECTION

BY

ALBERT LEFFINGWELL, M. D.

NEW YORK:
JOHN W. LOVELL COMPANY,
14 AND 16 VESEY STREET.

TO
A Memory of Friendship.

PREFACE.

To the Century Company of New York, in the pages of whose magazine, then known as "*Scribner's Monthly*," the first of the following essays originally appeared in July, 1880, the thanks of the writer are due for permission to re-publish in the present form. For a like courtesy on the part of the proprietors of Lippincott's Magazine, in which the second paper was first published [Aug., 1884], the writer desires to make due acknowledgment.

INTRODUCTION.

The first of the Essays following appeared in "SCRIBNER'S MONTHLY," in July, 1880; and immediately became honored by the attention of the Medical Press throughout the country. The aggressive title of the paper, justified, in great measure, perhaps, the vigor of the criticism bestowed. Again and again the point was raised by reviewers that the problem presented by the title, was not solved or answered by the article itself.

At this day, it perhaps may be mentioned that the question—"Does Vivisection Pay?" was never raised by the writer, who selected as his title the single word "Vivisection." The more taking headline was affixed by the editor of the magazine as more apt to arrest attention and arouse professional pugnacity. That in this latter respect it was eminently successful, the author had the best reason to remember. With this explanation—which is made simply to prevent future criticism on the same point—the old title is retained. If the present reader continues the inquiry here presented, he will learn wherein the writer believes in the utility of vivisection, and on the other hand, in what respects and under what conditions he very seriously questions whether any gains can possibly compensate the infinitely great cost.

"What do you hope for or expect as the result of agitation in regard to vivisection?" recently inquired a friend; "its legal abolition?"

"Certainly not," was the reply.

"Would you then expect its restriction during the present century?"

"Hardly even so soon as that. It will take longer than a dozen years to awaken recognition of any evil which touches neither the purse nor personal comfort of an American citizen. All that can be hoped in the immediate future is education. Action will perhaps follow when its necessity is recognized generally; but not before."

For myself, I believe no permanent or effective reform of present practices is probable until the Medical Profession generally concede as dangerous and unnecessary that freedom of unlimited experimentation in pain, which is claimed and practiced to-day. That legislative reform is otherwise unattainable, one would hesitate to affirm; but it assuredly would be vastly less effective. You must convince men of the justice and reasonableness of a law before you can secure a willing obedience. Yielding to none in loyalty to the science, and enthusiasm for the Art of Healing, what standpoint may be taken by those of the Medical Profession who desire to reform evils which confessedly exist?

I. We need not seek the total abolition of all experiments upon living animals. I do not forget that just such abolition is energetically demanded by a large number of earnest men and women, who have lost all faith in the possibility of restricting an abuse, if it be favored by scientific enthusiasm. "Let us take," they say, "the upright and conscientious ground of refusing all compromise with sin and evil, and maintaining our position unflinchingly, leave the rest to God."[1] This is almost precisely the ground taken by the Prohibitionists in national politics; it is the only ground one can occupy, provided the taking of a glass of wine, or the performance of any experiment,—painless or otherwise,—is of itself an "evil and a sin." There are those, however, who believe it possible to oppose and restrain intemperance by other methods than legislative prohibition. So with the prohibition of vivisection. Admitting the abuses of the practice, I cannot yet see that they are so intrinsic and essential as to make necessary the entire abolition of all physiological experiments whatsoever.

II. We may advocate (and I believe we should advocate)—*the total abolition, by law, of all mutilating or destructive experiments upon lower animals, involving pain, when such experiments are made for the purpose of public or private demonstration of already known and accepted physiological facts*.

This is the ground of compromise—unacceptable, as yet, to either party. Nevertheless it is asking simply for those limitations and restrictions which have always been conceded as prudent and fair by the medical profession of Great Britain. Speaking of a certain experiment upon the spinal nerves, Dr. M. Foster, of Cambridge University, one of the leading physiological teachers of England, says: "I have not performed it and have never seen it done," partly because of horror at the pain necessary. And yet this experiment has been performed before classes of young men and young women in the Medical Schools of this country! Absolutely no legal restriction here exists to the repetition, over and over again, of the most atrocious tortures of Mantegazza, Bert and Schiff.

This is the vivisection which does not "pay,"—even if we dismiss altogether from our calculation the interests of the animals sacrificed to the demand for mnemonic aid. For the great and perilous outcome of such methods will be—finally—an atrophy of the sense of sympathy for human suffering. It is seen to-day in certain hospitals in Europe. Can other result be expected to follow the deliberate infliction of prolonged pain without other object than to see or demonstrate what will happen therefrom? Will any assistance to memory, counterweigh the annihilation or benumbing of the instinct of pity?

Upon this subject of utility of painful experiments in class demonstrations or private study, I would like to appeal for judgment to the physician of the future, who then shall review the experience of the medical student of to-day. In his course of physiological training, he or she may be invited to see living animals cut and mutilated in various ways, eviscerated, poisoned, frozen, starved, and by ingenious devices of science subjected to the exhibition of pain. On the first occasion such a scene generally induces in the young man or young woman a significant subjective phenomenon of physiological interest; an involuntary, creeping, tremulous sense of horror emerges into consciousness,—and is speedily repressed. "This feeling," he whispers to himself, "is altogether unworthy the scientific spirit in which I am now to be educated; it needs to be subdued. The sight of this inarticulate agony, this prolonged anguish is not presented to me for amusement. I must steel myself to witness it, to assist in it, for the sake of the good I shall be helped thereby to accomplish, some day, for suffering humanity."

Praiseworthy sentiments, these are, indeed. Are they founded in reality? No. The student who thus conquers "squeamishness" will not see one fact thus demonstrated at the cost of pain which was unknown to science before; not one fact which he might not have been made to remember without this demonstrative illustration; *not one fact*—saddest truth of all—that is likely to be of the slightest practical service to him or to her in the multiplied and various duties of future professional life. Why, then, are they shown? To help him to remember his lesson! Admit the value to the student, but what of the cost?

In one of the great cities of China, I was shown, leaning against the high wall of the execution ground, a rude, wooden frame-work or cross, old, hacked, and smeared with recent blood-stains. It was used, I was told, in the punishment of extreme offenses; the criminal being bound thereto, and flayed and cut in every way human ingenuity could devise for inflicting torture before giving an immediately mortal wound. Only the week before, such an execution had taken place; the victim being a woman who had poisoned her husband. A young and enthusiastic physician whom I met, told me he had secured the privilege of being an eye witness to the awful tragedy, that he might verify a theory he had formed on the influence of pain; a theory perhaps like that which led to Mantegazza's crucifixion of pregnant rabbits with *dolori atrocissimi*.[2] Science here caught her profit from the punishment of crime, but the gain would have been the same had her interest alone been the object. There is *always* gain, always some aid to memory;—*but what of the cost?*

It cannot be expected that any Medical College, of its own accord and without outside pressure, will restrict or hamper its freedom of action. As a condition of prosperity and success it cannot show less than is exhibited by other medical schools; it must keep abreast of "advanced thought," and do and demonstrate in every way what its rivals demonstrate and do. There can be no question but that there is to-day a strong public demand for continental methods of physiological instruction. Who make this demand? You, gentlemen, students of medicine, and they who follow in your pathway. This year it is you who silently request this aid to your memory of the physiological statements of your text books; another year, another class of young men and young women, occupying the same benches, or filling the same laboratory, repeats the demand for the same series of illustrations. You, perhaps, will have gone forward to take your places in active life, to assume the real burdens of the medical profession. To those succeeding years of thought, reflection and usefulness, let me appeal, respecting the absolute necessity of all class demonstrations and laboratory work involving pain. Postpone if you please, the ready decision which, fresh from your class-room, you are perhaps only too willing to give me to-day; I do not wish it. But some time in the future, after years have gone by, remembering all you have seen and aided in the doing, tell us if you can, exactly wherein you received, in added potency for helping human suffering and for the treatment of human ills, the equivalent of that awful expenditure of pain which you are now demanding, and which by unprotesting acquiescence, you are *to-day* helping to inflict.

BOSTON, MASS.,
March, 1889.

[*From* SCRIBNER'S MONTHLY, *July, 1880*.]

DOES VIVISECTION PAY?

The question of vivisection is again pushing itself to the front. A distinguished American physiologist has lately come forward in defense of the French experimenter, Magendie, and, parenthetically, of his methods of investigation in the study of vital phenomena. On the other hand, the Society for the Prevention of Cruelty to Animals made an unsuccessful attempt, in the New York Legislature last winter, to secure the passage of a law which would entirely abolish the practice as now in vogue in our medical schools, or cause it to be secretly carried on, in defiance of legal enactments. In support of this bill it was claimed that physiologists, for the sake of "demonstrating to medical students certain physiological phenomena connected with the functions of life, are constantly and habitually in the practice of cutting up alive, torturing and tormenting divers of the unoffending brute creation to illustrate their theories and lectures, but without any practical or beneficial result either to themselves or to the students, which practice is demoralizing to both and engenders in the future medical practitioners a want of humanity and sympathy for physical pain and suffering." How far these statements are true will be hereafter discussed; but one assertion is so evidently erroneous that it may be at once indicated. *No* experiment, however atrocious, cruel and, therefore, on the whole, unjustifiable, if performed to illustrate some scientific point, was ever without "any beneficial result." The benefit may have been infinitesimal, but every scientific fact is of some value. To assert the contrary is to weaken one's case by overstatement.

Leaving out the brute creation, there are three parties interested in this discussion. In the first place, there are the professors and teachers of physiology in the medical colleges. Naturally, these desire no interference with either their work or their methods. They claim that were the knowledge acquired by experiments upon living organisms swept out of existence, in many respects the science of physiology would be little more than guesswork to-day. The subject of vivisection, they declare, is one which does not concern the general public, but belongs exclusively to scientists and especially to physiologists. That the present century should permit sentimentalists to interfere with scientific investigations is preposterous.

Behind these stand the majority of men belonging to the medical profession. Holding, as they do, the most important and intimate relations to society, it is manifestly desirable that they should enjoy the best facilities for the acquirement of knowledge necessary to their art. To most, the question is merely one of professional privilege against sentiment, and they cannot hesitate which side to prefer. In this, as in other professions or trades, the feeling of *esprit de corps* is exceedingly strong; and no class of men likes interference on the part of outsiders. To most physicians it is wholly a scientific question. It is a matter, they think, with which the public has no concern; if society can trust to the profession its sick and dying, they surely can leave to its feeling of humanity a few worthless brutes.

The opinion of the general public is therefore, divided and confused. On the one hand, it is profoundly desirous to make systematic and needless cruelty impossible; yet, on the other, it cannot but hesitate to take any step which shall hinder medical education, impede scientific discovery, or restrict search for new methods of treating disease. What are the sufferings of an animal, however acute or prolonged, compared with the gain to humanity which would result from the knowledge thereby acquired of a single curative agent? Public opinion hesitates. A leading newspaper, commenting on the introduction of the Bergh bill, doubtless expressed the sentiment of most people when it deprecated prevention of experiments "by which original investigators seek to establish or verify conclusions which may be of priceless value to the preservation of life and health among human beings."

The question nevertheless confronts society,—and in such shape, too, that society cannot escape, even if it would, the responsibility of a decision. Either by action or inaction the State must decide whether the practice of vivisection shall be wholly abolished, as desired by some; whether it shall be restricted by law within certain limits and for certain definite objects, as in Great Britain; or whether we are to continue in this country to follow the example of France and Germany, in permitting the practice of physiological experimentation to any extent devised or desired by the experimentalist himself. Any information tending to indicate which of these courses is best cannot be inopportune. Having witnessed experiments by some of the most distinguished European physiologists, such as Claude Bernard (the successor of Magendie), Milne-Edwards and Brown-Sequard; and, still better (or worse, as the reader may think), having performed some experiments in this direction for purposes of investigation and for the instruction of others, the present writer believes himself justified in holding and stating a pronounced opinion on this subject, even if it be to some extent, opposed to the one prevailing in the profession. Suppose, therefore, we review briefly the arguments to be adduced both in favor of the practice and against it.

Two principal arguments may be advanced in its favor.

I. It is undeniable that to the practice of vivisection we are indebted for very much of our present knowledge of physiology. This is the fortress of the advocates of vivisection, and a certain refuge when other arguments are of no avail.

II. As a means of teaching physiological facts, vivisection is unsurpassed. No teacher of science needs to be told the vast superiority of demonstration over affirmation. Take for instance, the circulation of the blood. The student who displays but a languid interest in statements of fact, or even in the best delineations and charts obtainable, will be thoroughly aroused by seeing the process actually before his eyes. A week's study upon the book will less certainly be retained in his memory than a single view of the opened thorax of a frog or dog. There before him is the throbbing heart; he sees its relations to adjoining structures, and marks, with a wonder he never before knew, that mystery of life by which the heart, even though excised from the body, does not cease for a time its rhythmic beat. To imagine, then, that teachers of physiology find mere amusement in these operations is the greatest of ignorant mistakes. They deem it desirable that certain facts be accurately fixed in memory, and they know that no system of mnemonics equals for such purpose the demonstration of the function itself.

Just here, however, arises a very important question. Admitting the benefit of the demonstration of scientific facts, *how far may one justifiably subject an animal to pain for the purpose of illustrating a point already known*? It is merely a question of cost. For instance, it is an undisputed statement in physical science that the diamond is nothing more than a form of crystallized carbon, and, like other forms of carbon, under certain conditions, may be made to burn. Now most of us are entirely willing to accept this, as we do the majority of truths, upon the testimony of scientific men, without making demonstration a requisite of assent. In a certain private school, however, it has long been the custom once a year, to burn in oxygen a small diamond, worth perhaps $30, so as actually to prove to the pupils the assertion of their text-books. The experiment is a brilliant one; no one can doubt its entire success. Nevertheless, we do not furnish diamonds to our public schools for this purpose. Exactly similar to this is one aspect of vivisection—it is a question of cost. Granting all the advantages which follow demonstration of certain physiological facts, the cost is pain—pain sometimes amounting to prolonged and excruciating torture. Is the gain worth this?

Let me mention an instance. Not long ago, in a certain medical college in the State of New York, I saw what Doctor Sharpey, for thirty years the professor of physiology in the University Medical College, London, once characterized by antithesis as "Magendie's *infamous* experiment," it having been first performed by that eminent physiologist. It was designed to prove that the stomach, although supplied with muscular coats, is during the act of vomiting for the most part passive; and that expulsion of its contents is due to the action of the diaphragm and the larger abdominal muscles. The professor to whom I refer did not propose to have even Magendie's word accepted as an authority on the subject: the fact should be demonstrated again. So an incision in the abdomen of a dog was made; its stomach was cut out; a pig's bladder containing colored water was inserted in its place, an emetic was injected into the veins,—and vomiting ensued. Long before the conclusion of the experiment the animal became conscious, and its cries of suffering were exceedingly painful to hear. Now, granting that this experiment impressed an abstract scientific fact upon the memories of all who saw it, nevertheless it remains significantly true that the fact thus demonstrated had no conceivable relation to the treatment of disease. It is not to-day regarded as conclusive of the theory which, after nearly two hundred repetitions of his experiment, was doubtless considered by Magendie as established beyond question. Doctor Sharpey, a strong advocate of vivisection, by the way, condemned it as a perfectly unjustifiable experiment, since "besides its atrocity, it was really purposeless." Was this repetition of the experiment which I have described worth its cost? Was the gain worth the pain?

Let me instance another and more recent case. Being in Paris a year ago, I went one morning to the College de France, to hear Brown-Sequard, the most eminent experimenter in vivisection now living—one who, Doctor Carpenter tells us, has probably inflicted more animal suffering than any other man in his time. The lecturer stated that injury to certain nervous centers near the base of the brain would produce peculiar and curious phenomena in the animal operated upon, causing it, for example, to keep turning to one side in a circular manner, instead of walking in a straightforward direction. A Guinea-pig was produced—a little creature, about the size of a half-grown kitten—and the operation was effected, accompanied by a series of piercing little squeaks. As foretold, the creature thus injured did immediately perform a "circular" movement. A rabbit was then operated upon with similar results. Lastly, an unfortunate poodle was introduced, its muzzle tied with stout whip-cord, wound round and round so tightly that it must necessarily have caused severe pain. It was forced to walk back and forth on the long table, during which it cast looks on every side, as though seeking a possible avenue of escape. Being fastened in the operating trough, an incision was made to the bone, flaps turned back, an opening made in the skull, and enlarged by breaking away some portions with forceps. During these various processes no attempt whatever was made to cause unconsciousness by means of anæsthetics, and the half-articulate, half-smothered cries of the creature in its agony were terrible to hear, even to one not unaccustomed to vivisections. The experiment was a "success"; the animal after its mutilation *did* describe certain circular movements. But I cannot help questioning in regard to these demonstrations, *did they pay*? This experiment had not the slightest relation whatever to the cure of disease. More than this: it teaches us little or nothing in physiology. The most eminent physiologist in this country, Doctor Austin Flint, Jr., admits that experiments of this kind "do not seem to have advanced our positive knowledge of the functions of the nerve centers," and that similar experiments "have been very indefinite in their results." On this occasion, therefore, three animals were subjected to torture to demonstrate an abstract fact, which probably not a single one of the two dozen spectators would have hesitated to take for granted on the word of so great a pathologist as Doctor Brown-Sequard. Was the gain worth the cost?

This, then, is the great question that must eventually be decided by the public. Do humanity and science here indicate diverging roads? On the contrary, I believe it to be an undeniable fact that *the highest scientific and medical opinion is against the repetition of painful experiments for class teaching*. In 1875, a Royal Commission was appointed in Great Britain to investigate the subject of vivisection, with a view to subsequent legislation. The interests of science were represented by the appointment of Professor Huxley as a member of this commission. Its meetings continued over several months, and the report constitutes a large volume of valuable testimony. The opinions of many of these witnesses are worthy of special attention, from the eminent position to the men who hold them. The physician to the Queen, Sir Thomas Watson, with whose "Lectures on Physic" every medical practitioner in this country is familiar, says: "I hold that no teacher or man of science who by his own previous experiments, * * * has thoroughly satisfied himself of the solution of any physiological problem, is justified in repeating the experiments, however mercifully, to appease the natural curiosity of a class of students or of scientific friends." Sir George Burroughs, President of the Royal College of Physicians, says: "I do not think that an experiment should be repeated over and over again in our medical schools to illustrate what is already established."[3] Sir James Paget, Surgeon Extraordinary to the Queen, said before the commission that "experiments for the purpose of repeating anything already ascertained ought never to be shown to classes." [363.] Sir William Fergusson, F. R. S., also Surgeon to her Majesty, asserted that "sufferings incidental to such operations are protracted in a very shocking manner"; that of such experiments there is "useless repetition," and that "when once a fact which involves cruelty to animals has been fairly recognized and accepted, there is no necessity for a continued repetition." [1019.] Even physiologists—some of them practical experimenters in vivisection—join in condemning these class demonstrations. Dr. William Sharpey, before referred to as a teacher of physiology for over thirty years in University College, says: "Once such facts fully established, I do not think it justifiable to repeat experiments causing pain to animals." [405.] Dr. Rolleston, Professor of Physiology at Oxford, said that "for class demonstrations limitations should undoubtedly be imposed, and *those limitations should render illegal painful experiments before classes*." [1291.] Charles Darwin, the greatest of living naturalists, stated that he had never either directly or indirectly experimented on animals, and that he regarded a painful experiment without anæsthetics which might be made with anæsthetics as deserving "detestation and abhorrence." [4672.] And finally the report of this commission, to which is attached the name of Professor Huxley, says: "With respect to medical schools, we accept the resolution of the British Association in 1871, that experimentation without the use of anæsthetics is not a fitting exhibition for teaching purposes."

It must be noted that hardly any of these opinions touch the question of vivisection so far as it is done without the infliction of pain, nor object to it as a method of original research; they relate simply to the practice of repeating painful experiments for purposes of physiological teaching. We cannot dismiss them as "sentimental" or unimportant. If painful experiments are necessary for the education of the young physician, how happens it that Watson and Burroughs are ignorant of the fact? If indispensable to the proper training of the surgeon, why are they condemned by Fergusson and Paget? If requisite even to physiology, why denounced by the physiologists of Oxford and London? If necessary to science, why viewed "with abhorrence" by the greatest of modern scientists?

Another objection to vivisection, when practiced as at present without supervision or control, is the undeniable fact that habitual familiarity with the infliction of pain upon animals has a decided tendency to engender a sort of careless indifference regarding suffering. "Vivisection," says Professor Rolleston of Oxford, "is very liable to abuse. * * * It is specially liable to tempt a man into certain carelessness; the passive impressions produced by the sight of suffering growing weaker, while the habit and pleasure of experimenting grows stronger by repetition." [1287.] Says Doctor Elliotson: "I cannot refrain from expressing my horror at the amount of torture which Doctor Brachet inflicted. *I hardly think knowledge is worth having at such a purchase.*"[4] A very striking example of this tendency was brought out in the testimony of a witness before the Royal Commission,—Doctor Klein, a practical physiologist. He admitted frankly that as an investigator he held as entirely indifferent the sufferings of animals subjected to his experiments, that, except for teaching purposes, he never used anæsthetics unless necessary for his own convenience. Some members of the Commission could hardly realize the possibility of such a confession.

"Do you mean you have no regard at all to the sufferings of the lower animals?"

"*No regard at all,*" was the strange reply; and, after a little further questioning, the witness explained:

"I think that, with regard to an experimenter—a man who conducts special research and performs an experiment—he has *no time, so to speak, for thinking what the animal will feel or suffer*!"

Of Magendie's cruel disposition there seems only too abundant evidence. Says Doctor Elliotson: "Dr. Magendie, in one of his barbarous experiments, which I am ashamed to say I witnessed, began by coolly cutting out a large round piece from the back of a beautiful little puppy, as he would from an apple dumpling!" "It is not to be doubted that inhumanity may be found in persons of very high position as physiologists. *We have seen that it was so in Magendie.*" This is the language of the report on vivisection, to which is attached the name of Professor Huxley.

But the fact which, in my own mind, constitutes by far the strongest objection to unrestrained experiments in pain, is their questionable utility as regards therapeutics. Probably most readers are aware that physiology is that science which treats of the various functions of life, such as digestion, respiration and the circulation of the blood, while therapeutics is that department of medicine which relates to the discovery and application of remedies for disease. Now I venture to assert that, during the last quarter of a century, infliction of intense torture upon unknown myriads of sentient, living creatures, *has not resulted in the discovery of a single remedy of acknowledged and generally accepted value in the cure of disease*. This is not known to the general public, but it is a fact essential to any just decision regarding the expediency of unrestrained liberty of vivisection. It is by no means intended to deny the value to therapeutics of well-known physiological facts acquired thus in the past—such, for instance, as the more complete knowledge we possess regarding the circulation of the blood, or the distinction between motor and sensory nerves, nor can original investigation be pronounced absolutely valueless as respects remote possibility of future gain. What the public has a right to ask of those who would indefinitely prolong these experiments without State supervision or control is, "What good have your painful experiments accomplished during the past thirty years—not in ascertaining facts in physiology or causes of rare or incurable complaints, but in the discovery of improved methods for ameliorating human suffering, and for the cure of disease?" If pain could be estimated in money, no corporation ever existed which would be satisfied with such waste of capital in experiments so futile; no mining company would permit a quarter-century of "prospecting" in such barren regions. The usual answer to this inquiry is to bring forward facts in physiology thus acquired in the past, in place of facts in therapeutics. Thus, in a recent article on Magendie to which reference has been made, we are furnished with a long list of such additions to our knowledge. It may be questioned, however, whether the writer is quite scientifically accurate in asserting that, were our past experience in vivisection abolished, "it would blot out *all* that we know to-day in regard to the circulation of the blood, * * the growth and regeneration of bone, * * * the origin of many parasitic diseases, * * * the communicability of certain contagious and infectious diseases, and, to make the list complete, it would be requisite * * to take *a wide range in addition through the domains of pathology and therapeutics*." Surely somewhat about these subjects has been acquired otherwise than by experiments upon animals? For example, an inquiring critic might wish to know a few of the "many parasitic diseases" thus discovered; or what contagious and infectious diseases, whose communicability was previously unknown, have had this quality demonstrated solely by experiments on animals? And what, too, prevented that "wide range into therapeutics" necessary to make complete the list of benefits due to vivisection? In urging the utility of a practice so fraught with danger, the utmost precaution against the slightest error of overstatement becomes an imperative duty. Even so distinguished a scientist as Sir John Lubbock once rashly asserted in Parliament that, "without experiments on living animals, we should never have had the use of ether"! Nearly every American school-boy knows that the contrary is true—that the use of ether as an anæsthetic—the grandest discovery of modern times—had no origin in the torture of animals.

I confess that, until very recently, I shared the common impression regarding the utility of vivisection in therapeutics. It is a belief still widely prevalent in the medical profession. Nevertheless, is it not a mistake? The therapeutical results of nearly half a century of painful experiments—we seek them in vain. Do we ask surgery? Sir William Ferguson, surgeon to the Queen, tells us: "In surgery I am not aware of any of these experiments on the lower animals having led to the mitigation of pain or to improvement as regards surgical details." [1049.] Have antidotes to poisons been discovered thereby? Says Doctor Taylor, lecturer on Toxicology for nearly half a century in the chief London Medical School (a writer whose work on Poisons is a recognized authority): "I do not know that we have as yet learned anything, so far as treatment is concerned, from our experiments with them (*i.e.* poisons) on animals." [1204.] Doctor Anthony, speaking of Magendie's experiments, says: "I never gained one single fact by seeing these cruel experiments in Paris. *I know nothing more from them than I could have read.*" [2450.] Even physiologists admit the paucity of therapeutic results. Doctor Sharpey says: "I should lay less stress on the direct application of the results of vivisection to improvement in the art of healing, than upon the value of these experiments in the promotion of physiology." [394.] The Oxford professor of Physiology admitted that Etiology, the science which treats of the causes of disease, had, by these experiments, been the gainer, rather than therapeutics. [1302.] "Experiments on animals," says Doctor Thorowgood, "already extensive and numerous, cannot be said to have advanced therapeutics much."[5] Sir William Gull, M. D., was questioned before the commission whether he could enumerate any therapeutic remedies which have been discovered by vivisection, and he replied with fervor: "The cases bristle around us everywhere!" Yet, excepting Hall's experiments on the nervous system, he could enumerate only various forms of disease, our knowledge of which is due to Harvey's discovery, two hundred and fifty years ago! The question was pushed closer, and so brought to the necessity of a definite reply, he answered: "I do not say at present our therapeutics are much, but there are lines of experiment which *seem to promise* great help in therapeutics." [5529.] The results of two centuries of experiments, so far as therapeutics are concerned, reduced to a seeming promise!

On two points, then, the evidence of the highest scientific authorities in Great Britain seems conclusive—first, that experiments upon living animals conduce chiefly to the benefit of the science of physiology, and little, if at all, at the present day, to the treatment of disease or the amelioration of human suffering; and, secondly, that repetition of painful experiments for class-teaching in medical schools is both unnecessary and unjustifiable. Do these conclusions affect the practice of vivisection in this country? Is it true that experiments are habitually performed in some of our medical schools, often causing extreme pain, to illustrate well-known and accepted facts—experiments which English physiologists pronounce "infamous" and "atrocious," which English physicians and surgeons stigmatize as purposeless cruelty and unjustifiable—which even Huxley regards as unfitting for teaching purposes, and Darwin denounces as worthy of detestation and abhorrence? I confess I see no occasion for any over-delicate reticence in this matter. Science needs no secrecy either for her methods or results; her function is to reveal, not to hide, facts. The reply to these questions must be in the affirmative. In this country our physiologists are rather followers of Magendie and Bernard, after the methods in vogue at Paris and Leipsic, than governed by the cautious and sensitive conservatism in this respect which generally characterizes the physiological teaching of London and Oxford. In making this statement, no criticism is intended on the motives of those responsible for ingrafting continental methods upon our medical schools. If any opprobrium shall be inferred for the past performance of experiments herein condemned, the present writer asks a share in it. It is the future that we hope to change. Now, what are the facts? A recent contributor to the "International Review," referring to Mr. Bergh, says that "he assails physiological experiments with the same blind extravagance of denunciation as if they were still performed without anæsthetics, as in the time of Magendie." In the interests of scientific accuracy one would wish more care had been given to the construction of this sentence, for it implies that experiments are not now performed except with anæsthetics—a meaning its author never could have intended to convey. Every medical student in New York knows that experiments involving pain are repeatedly performed to illustrate teaching. It is no secret; one need not go beyond the frank admissions of our later text-books on physiology for abundant proof, not only of this, but of the extent to which experimentation is now carried in this country. "We have long been in the habit, in class demonstrations, of removing the optic lobe on one side from a pigeon," says Professor Flint, of Bellevue Hospital Medical College, in his excellent work on Physiology.[6] "The experiment of dividing the sympathetic in the neck, especially in rabbits, is so easily performed that the phenomena observed by Bernard and Brown-Sequard have been repeatedly verified. *We have often done this in class demonstrations.*"[7] "The cerebral lobes were removed from a young pigeon in the usual way, an operation * * *which we practice yearly as a class demonstration*."[8] Referring to the removal of the cerebellum, the same authority states: "Our own experiments, which have been very numerous during the last fifteen years, are *simply repetitions of those of Flourens, and the results have been the same without exception."*[9] *We have frequently removed both kidneys* from dogs, and when the operation is carefully performed the animals live for from three to five days. * * Death always takes place with symptoms of blood poisoning."[10] In the same work we are given precise details for making a pancreatic fistula, after the method of Claude Bernard—"one we have repeatedly employed with success." "In performing the above experiment it is generally better *not* to employ an anæsthetic,"[11] but ether is sometimes used. In the same work is given a picture of a dog, muzzled and with a biliary fistula, as it appeared the fourteenth day after the operation, which, with details of the experiment, is quite suggestive.[12] Bernard was the first to succeed in following the spinal accessory nerve back to the jugular foramen, seizing it here with a strong pair of forceps and drawing it out by the roots. This experiment is practiced in our own country. "We have found this result (loss of voice) to follow in the cat after the spinal accessory nerves have been torn out by the roots," says Professor John C. Dalton, in his Treatise on Human Physiology.[13] "This operation is difficult," writes Professor Flint, "but we have several times performed it with entire success;" and his assistant at Bellevue Medical College has succeeded "in extirpating these nerves for class demonstrations."[14] In withdrawal of blood from the hepatic veins of a dog, "avoiding the administration of an anæsthetic" is one of the steps recommended.[15] The curious experiment of Bernard, in which artificial diabetes is produced by irritating the floor of the fourth ventricle of the brain, is carefully described, and illustrations afforded both of the instrument and the animal undergoing the operation. The inexperienced experimenter is here taught to hold the head of the rabbit "firmly in the left hand," and to bore through its skull "by a few lateral movements of the instrument." It is not a difficult operation; it is one which the author has "often repeated." He tell us "*it is not desirable* *to administer an anæsthetic*," as it would prevent success; and a little further we are told that "we should avoid the administration of anæsthetics in all accurate experiments on the glycogenic function."[16] It is true the pleasing assurance is given that "this experiment is almost painless"; but on this point, could the rabbit speak during the operation, its opinion might not accord with that of the physiologist.

There is one experiment in regard to which the severe characterization of English scientists is especially applicable, from the pain necessarily attending it. Numerous investigators have long established the fact that the great sensory nerve of the head and face is endowed with an exquisite degree of sensibility. More than half a century ago, both Magendie and Sir Charles Bell pointed out that merely exposing and touching this fifth nerve gave signs of most acute pain. "All who have divided this root in living animals must have recognized, not only that it is sensitive, but that its sensibility is far more acute than that of any other nervous trunk in the body."[17] "The fifth pair," says Professor John C. Dalton, "is the most acutely sensitive nerve in the whole body. Its irritation by mechanical means *always causes intense pain*, and even though the animal be nearly unconscious from the influence of ether, any severe injury to its large root is almost invariably followed by cries."[18] Testimony on this point is uniform and abundant. If science speaks anywhere with assurance, it is in regard to the properties of this nerve. Yet every year the experiment is repeated before medical classes, simply to demonstrate accepted facts. "This is an operation," says Professor Flint, referring to the division of this nerve, "that we have frequently performed with success." He adds that "it is difficult from the fact that one is working in the dark, and it requires a certain amount of dexterity, *to be acquired only by practice*." Minute directions are therefore laid down for the operative procedure, and illustrations given both of the instrument to be used, and of the head of a rabbit with the blade of the instrument in its cranial cavity.[19] Holding the head of our rabbit firmly in the left hand, we are directed to penetrate the cranium in a particular manner. "Soon the operator feels at a certain depth that the bony resistance ceases; he is then on the fifth pair, and the cries of the animal give evidence that the nerve is pressed upon." This is one of Magendie's celebrated experiments; perhaps the reader fancies that in its modern repetitions the animal suffers nothing, being rendered insensible by anæsthetics? "*It is* *much more satisfactory to divide the nerve without etherizing the animal, as the evidence of pain is an important guide in this delicate operation.*" Anæsthetics, however, are sometimes used, but not so as wholly to overcome the pain.

Testimony of individuals, indicating the extent to which vivisection is at present practiced in this country might be given; but it seems better to submit proof within the reach of every reader, and the accuracy of which is beyond cavil. No legal restrictions whatever exist, preventing the performance of any experiment desired. Indeed, I think it may safely be asserted that, in the city of New York, in a single medical school, more pain is inflicted upon living animals as a means of teaching well-known facts, than is permitted to be done for the same purpose in all the medical schools of Great Britain and Ireland. And *cui bono*? "I can truly say," writes a physician who has seen all these experiments, "that not only have I never seen any results at all commensurate with the suffering inflicted, but I cannot recall a single experiment which, in the slightest degree, has increased my ability to relieve pain, or in any way fitted me to cope better with disease."

In respect to this practice, therefore, evidence abounds indicating the necessity for that State supervision which obtains in Great Britain. We cannot abolish it any more than we can repress dissection; to attempt it would be equally unwise. Within certain limitations, dictated both by a regard for the interest of science and by that sympathy for everything that lives and suffers which is the highest attribute of humanity, it seems to me that the practice of vivisection should be allowed. What are these restrictions?

The following conclusions are suggested as a basis for future legislation:

I. Any experiment or operation whatever upon a living animal, during which by recognized anæsthetics it is made completely insensible to pain, should be permitted.

This does not necessarily imply the taking of life. Should a surgeon, for example, desire to cause a fracture or tie an artery, and then permit the animal to recover so as to note subsequent effects, there is no reason why the privilege should be refused. The discomfort following such an operation would be inconsiderable. This permission should not extend to experiments purely physiological and having no definite relation to surgery; nor to mutilation from which recovery is impossible, and prolonged pain certain as a sequence.

II. Any experiment performed thus, under complete anæsthesia, though involving any degree of mutilation, if concluded by the extinction of life before consciousness is regained should also be permitted.

To object to killing animals for scientific purposes while we continue to demand their sacrifice for food, is to seek for the appetite a privilege we refuse the mind. It is equally absurd to object to vivisection because it dissects, or "cuts up." If no pain be felt, why is it worse to cut up a dog, than a sheep or an ox? Such experiments as the foregoing might be permitted to any extent desired in our medical schools.

Far more difficult is the question of painful experimentation. Unfortunately, it so happens that the most attractive original investigations are largely upon the nervous system, involving the consciousness of pain as a requisite to success. Toward this class of experiments the State should act with caution and firmness. It seems to me that the following restrictions are only just.

III. In view of the great cost in suffering, as compared with the slight profit gained by the student, the repetition, for purposes of class instruction of any experiment involving pain to a vertebrate animal should be forbidden by law.

IV. In view of the slight gain to practical medicine resulting from innumerable past experiments of this kind, a painful experiment upon a living vertebrate animal should be permitted solely for purposes of original investigation, and then only under the most rigid surveillance, and preceded by the strictest precautions. For every experiment of this kind the physiologist should be required to obtain special permission from a State board, specifying on application (1) the object of the proposed investigation, (2) the nature and method of the operation, (3) the species of animal to be sacrificed, and (4) the shortest period during which pain will probably be felt. An officer of the State should be given an opportunity to be present; and a report made, both of the length of time occupied, and the knowledge, if any, gained thereby. If these restrictions are made obligatory by statute, and their violation made punishable by a heavy fine, such experiments will be generally performed only when absolutely necessary for purposes of scientific research.

In few matters is there greater necessity for careful discrimination than in everything pertaining to this subject. The attempt has been made in this paper to indicate how far the State—leaning to mercy's side—may sanction a practice often so necessary and useful, always so dangerous in its tendencies. That is a worthy ideal of conduct which seeks

> "Never to blend our pleasure or our pride
> With sorrow of the meanest thing that feels."

Is not this a sentiment in which even science may fitly share? Are we justified in neglecting the evidence she offers, purchased in the past at such immeasurable agonies, and in demanding that year after year new victims shall be subjected to torture, only to demonstrate what none of us doubt? That is the chief question. For, if all compromise be persistently rejected by physiologists, there is danger that some day, impelled by the advancing growth of humane sentiment, society may confound in one common condemnation all experiments of this nature, and make the whole practice impossible, except in secret and as a crime.

[*From* LIPPINCOTT'S MAGAZINE, *August, 1884*.]

VIVISECTION.

Omitting entirely any consideration of the ethics of vivisection, the only points to which in the present article the attention of the reader is invited are those in which scientific inquirers may be supposed to have a common interest.

I. One danger to which scientific truth seems to be exposed is a peculiar tendency to underestimate the numberless uncertainties and contradictions created by experimentation upon living beings. Judging from the enthusiasm of its advocates, one would think that by this method of interrogating nature all fallacies can be detected, all doubts determined. But, on the contrary, the result of experimentation, in many directions, is to plunge the observer into the abyss of uncertainty. Take, for example, one of the simplest and yet most important questions possible,—the degree of sensibility in the lower animals. Has an infinite number of experiments enabled physiologists to determine for us the mere question of pain? Suppose an amateur experimenter in London, desirous of performing some severe operations upon frogs, to hesitate because of the extreme painfulness of his methods, what replies would he be likely to obtain from the highest scientific authorities of England as to the sensibility of these creatures? We may fairly judge their probable answers to such inquiries from their evidence already given before a royal commission.[20]

Dr. Carpenter would doubtless repeat his opinion that "frogs have extremely little perception of pain;" and in the evidence of that experienced physiologist George Henry Lewes, he would find the cheerful assurance, "I do not believe that frogs suffer pain at all." Our friend applies, let us suppose, to Dr. Klein, of St. Bartholomew's Hospital, who despises the sentimentality which regards animal suffering as of the least consequence; and this enthusiastic vivisector informs him that, in his English experience, the experiment which caused the greatest pain without anæsthetics was the cauterization of the cornea of a frog. Somewhat confused at finding that a most painful experiment can be performed upon an animal that does not suffer he relates this to Dr. Swaine Taylor, of Guy's Hospital, who does not think that Klein's experiment would cause severe suffering; but of another—placing a frog in cold water and raising the temperature to about 100°—"that," says Doctor Taylor, "would be a cruel experiment: I cannot see what purpose it can answer." Before leaving Guy's Hospital, our inquiring friend meets Dr. Pavy, one of the most celebrated physiologists in England, who tells him that in this experiment, stigmatized by his colleague as "cruel," the frog would in reality suffer very little; that if we ourselves were treated to a bath gradually raised from a medium temperature to the boiling point, "I think we should not feel any pain;" that were we plunged at once into boiling water, "even then," says the enthusiastic and scientific Dr. Pavy, "I do not think pain would be experienced!" Our friend goes then to Dr. Sibson, of St. Mary's Hospital, who as a physiologist of many years' standing, sees no objection to freezing, starving, or baking animals alive; but he declares of boiling a frog, "That is a horrible idea, and I certainly am not going to defend it." Perplexed more than ever, he goes to Dr. Lister, of King's College, and is astonished upon being told "that the mere holding of a frog in your warm hand is about as painful as any experiment probably that you would perform." Finally, one of the strongest advocates of vivisections, Dr. Anthony, pupil of Sir Charles Bell, would exclaim, if a mere exposition of the lungs of the frog were referred to, "Fond as I am of physiology, I would not do that for the world!"

Now, what has our inquirer learned by his appeal to science? Has he gained any clear and absolute knowledge? Hardly two of the experimenters named agree upon one simple yet most important preliminary of research—*the sensibility to pain of a single species of animals*.

Let us interrogate scientific opinion a little further on this question of sensibility. Is there any difference in animals as regards susceptibility to pain? Dr. Anthony says that we may take the amount of intelligence in animals as a fair measure of their sensibility—that the pain one would suffer would be in proportion to its intelligence. Dr. Rutherford, Edinburgh, never performs an experiment upon a cat or a spaniel if he can help it, because they are so exceedingly sensitive; and Dr. Horatio Wood, of Philadelphia, tells us that the nervous system of a cat is far more sensitive than that of the rabbit. On the other hand, Dr. Lister, of King's College, is not aware of any such difference in sensibility in animals, and Dr. Brunton, of St. Bartholomew's, finds cats such very good animals to operate with that he on one occasion used ninety in making a single experiment.

Sir William Gull thinks "there are but few experiments performed on living creatures where sensation is not removed," yet Dr. Rutherford admits "about half" his experiments to have been made upon animals sensitive to pain. Professor Rolleston, of Oxford University, tells us "the whole question of anæsthetizing animals has an element of uncertainty"; and Professor Rutherford declares it "impossible to say" whether even artificial respiration is painful or not, "unless the animal can speak." Dr. Brunton, of St. Bartholomew's, says of that most painful experiment, poisoning by strychnine, that it cannot be efficiently shown if the animal be under chloroform. Dr. Davy, of Guy's, on the contrary, always gives chloroform, and finds it no impediment to successful demonstration, Is opium an anæsthetic? Claude Bernard declares that sensibility exists even though the animal be motionless: "*Il sent la douleur, mais il a, pour ainsi dire, perdu l'idee de la defense.*"[21] But Dr. Brunton, of St. Bartholomew's hospital, London, has no hesitation whatever in contradicting this statement "emphatically, however high an authority it may be."

Curare, a poison invented by South American Indians for their arrows, is much used in physiological laboratories to paralyze the motor nerves, rendering an animal absolutely incapable of the slightest disturbing movement. Does it at the same time destroy sensation, or is the creature conscious of every pang? Claude Bernard, of Paris, Sharpey, of London, and Flint, of New York[22] all agree that sensation is *not* abolished; on the other hand, Rutherford regards curare as a partial anæsthetic, and Huxley strongly intimates that Bernard in thus deciding from experiments that it does not affect the cerebral hemispheres or consciousness, "*jumped at a conclusion* for which neither he nor anybody else had any scientific justification." This is extraordinary language for one experimentalist to use regarding others! If it is possible that such men as Claude Bernard and Professor Flint have "jumped at" one utterly unscientific conclusion, notwithstanding the most painstaking of vivisections, what security have we that other of our theories in physiology now regarded as absolutely established may not be one day as severely ridiculed by succeeding investigators? Is it, after all, true, that the absolute certainty of our most important deductions must remain forever hidden "unless the animal can speak"?

II. Between advocating State supervision of painful vivisection, and proposing with Mr. Bergh the total suppression of all experiments, painful or otherwise, there is manifestly a very wide distinction. Unfortunately, the suggestion of any interference whatever invariably rouses the anger of those most interested—an indignation as unreasonable, to say the least, as that of the merchant who refuses a receipt for money just paid to him, on the ground that a request for a written acknowledgement is a reflection upon his honesty. I cannot see how otherwise than by State supervision we are to reach abuses which confessedly exist. Can we trust the sensitiveness and conscience of every experimenter? Nobody claims this. One of the leading physiologists in this country, Dr. John C. Dalton, admits "that vivisection may be, and has been, abused by reckless, unfeeling, or unskillful persons;" that he himself has witnessed abroad, in a veterinary institution, operations than which "nothing could be more shocking." And yet the unspeakable atrocities at Alfort, to which, apparently, Dr. Dalton alludes, were defended upon the very ground he occupies to-day in advocating experiments of the modern laboratory and classroom; for the Academie des Sciences decided that there was "no occasion to take any notice of complaints; that in the future, as in the past, vivisectional experiments must be left entirely to the judgment of scientific men." What seemed "atrocious" to the more tender-hearted Anglo-Saxon was pronounced entirely justifiable by the French Academy of Science.

A curious question suggests itself in connection with this point. There can be little doubt, I think, that the sentiment of compassion and of sympathy with suffering is more generally diffused among all classes of Great Britain than elsewhere in Europe; and one cannot help wondering what our place might be, were it possible to institute any reliable comparison of national humanity. Should we be found in all respects as sensitive as the English people? Would indignation and protest be as quickly and spontaneously evoked among us by a cruel act? The question may appear an ungracious one, yet it seems to me there exists some reason why it should be plainly asked. There is a certain experiment—one of the most excruciating that can be performed—which consists in exposing the spinal cord of the dog for the purpose of demonstrating the functions of the spinal nerves. It is one, by the way, which Dr. Wilder forgot to enumerate in his summary of the "four kinds of experiments," since it is not the "cutting operation" which forms its chief peculiarity or to which special objection would be made. At present all this preliminary process is generally performed under anæsthetics: it is an hour or two later, when the animal has partly recovered from the severe shock of the operation, that the wound is reopened and the experiment begins. It was during a class demonstration of this kind by Magendie, before the introduction of ether, that the circumstance occurred which one hesitates to think possible in a person retaining a single spark of humanity or pity. "I recall to mind," says Dr. Latour, who was present at the time, "a poor dog, the roots of whose vertebral nerves Magendie desired to lay bare to demonstrate Bell's theory, which he claimed as his own. The dog, mutilated and bleeding twice escaped from under the implacable knife, and threw its front paws around Magendie's neck, licking, as if to soften his murderer and ask for mercy! I confess I was unable to endure that heartrending spectacle."

It was probably in reference to this experiment that Sir Charles Bell, the greatest English physiologist of our century, writing to his brother in 1822, informs him that he hesitates to go on with his investigations. "You may think me silly," he adds, "but I cannot perfectly convince myself that I am authorized in nature or religion to do these cruelties." Now, what do English physiologists and vivisectors of the present day think of the repetition of this experiment solely as a class demonstration?

They have candidly expressed their opinions before a royal commission. Dr. David Ferrier, of King's college, noted for his experiments upon the brain of monkeys, affirms his belief that "students would rebel" at the sight of a painful experiment. Dr. Rutherford, who certainly dared do all that may become a physiologist, confesses mournfully, "*I dare not* show an experiment upon a dog or rabbit before students, when the animal is not anæsthetized." Dr. Pavy, of Guy's Hospital, asserts that a painful experiment introduced before a class "would not be tolerated for a moment." Sir William Gull, M. D., believes that the repetition of an operation like this upon the spinal nerves would excite the reprobation alike of teacher, pupils, and the public at large. Michael Foster, of Cambridge University, who minutely describes all the details of the experiment on recurrent sensibility in the "Handbook for the Physiological Laboratory," nevertheless tells us, "I have not performed it, and have never seen it done," partly, as he confesses, "from horror at the pain." And finally Dr. Burdon-Sanderson, physiologist at University College, London, states with the utmost emphasis, in regard to the performance of this demonstration on the spinal cord, "I am perfectly certain that no physiologist—none of the leading men in Germany, for example—would exhibit an experiment of that kind."

Now mark the contrast. This experiment—which we are told passes even the callousness of Germany to repeat; which every leading champion of vivisection in Great Britain reprobates for medical teaching; which some of them shrink even from seeing, themselves, from horror at the tortures necessarily inflicted; which the most ruthless among them *dare not* exhibit to the young men of England,—*this experiment has been performed publicly again and again in American medical colleges*, without exciting, so far as we know, even a whisper of protest or the faintest murmur of remonstrance! The proof is to be found in the published statements of the experimenter himself. In his "Text-Book of Physiology," Professor Flint says, "Magendie showed very satisfactorily that the posterior roots (of the spinal cord) were exclusively sensory, and this fact has been confirmed by more recent observations upon the higher classes of animals. We have ourselves frequently exposed and irritated the roots of the nerves in dogs, *in public demonstrations* in experiments on the recurrent sensibility, ... and in another series of observations."[23]

This is the experience of a single professional teacher; but it is improbable that this experiment has been shown only to the students of a single medical college in the United States; it has doubtless been repeated again and again in different colleges throughout the country. If Englishmen are, then, so extremely sensitive as Ferrier, Gull, and Burdon-Sanderson would have us believe, we must necessarily conclude that the sentiment of compassion is far greater in Britain than in America. Have we drifted backward in humanity? Have American students learned to witness, without protest, tortures at the sight of which English students would rebel? We are told that there is no need of any public sensitiveness on this subject. We should trust entirely, as they do in France,—at Alfort, for example,—"to the judgment of the investigator." There must be no lifting of the veil to the outside multitude; for the priests of this unpitying science there must be as absolute immunity from criticism or inquiry as was ever demanded before the shrine of Delphi or the altars of Baal. "Let them exercise their solemn office," demands Dr. Wilder, "not only unrestrained by law, but upheld by public sentiment."

For myself, I cannot believe this position is tenable. Nothing seems to me more certain than the results that must follow if popular sentiment in this country shall knowingly sustain the public demonstration of an experiments in pain, which can find no defender among the physiologists of Great Britain. It has been my fortune to know something of the large hospitals of Europe; and I confess I do not know a single one in countries where painful vivisection flourishes, unchecked by law, wherein the poor and needy sick are treated with the sympathy, the delicacy, or even the decency, which so universally characterize the hospitals of England. When Magendie, operating for cataract, plunged his needle to the bottom of his patient's eye, that he might note upon a human being the effect produced by mechanical irritation of the retina, he demonstrated how greatly the zeal of the enthusiast may impair the responsibility of the physician and the sympathy of man for man.

III. The utility of vivisection in advancing therapeutics, despite much argument, still remains an open question. No one is so foolish as to deny the possibility of future usefulness to any discovery whatever; but there is a distinction, very easily slurred over in the eagerness of debate, between present applicability and remotely potential service. If the pains inflicted on animals are absolutely necessary to the protection of human life and the advancement of practical skill in medicine, should sentiment be permitted to check investigation? An English prelate, the Bishop of Peterborough, speaking in Parliament on this subject, once told the House of Lords that "it was very difficult to decide what was unnecessary pain," and as an example of the perplexities which arose in his own mind he mentioned "the case of the wretched man who was convicted of skinning cats alive, because their skins were more valuable when taken from the living animal than from the dead one. The extra money," added the Bishop, "got the man a dinner!"[24] Whether in this particular case the excuse was well received by the judge, the reverend prelate neglected to inform us; but it is certain that the plea for painful experimentation rests substantially on the same basis. Out of the agonies of sentient brutes we are to pluck the secret of longer living and the art of surer triumph over intractable disease.

But has this hope been fulfilled? Pasteur, we are told, has claimed the discovery of a cure for hydrophobia through experiments on animals. It may be well worth its cost if only true; but we cannot forget that its practical value is by no means yet demonstrated. Aside from this, has physiological experimentation during the last quarter of a century contributed such marked improvements in therapeutic methods that we find certain and tangible evidence thereof in the diminishing fatality of any disease? Can one mention a single malady which thirty years ago resisted every remedial effort, to which the more enlightened science of to-day can offer hopes of recovery? These seem to me perfectly legitimate and fair questions, and, fortunately, in one respect, capable of a scientific reply. I suppose the opinion of the late Claude Bernard, of Paris, would be generally accepted as that of the highest scientific authority on the utility of vivisection in "practical medicine;" but he tells us that it is hardly worth while to make the inquiry. "Without doubt," he confessed, "*our hands are empty to-day*, although our mouths are full of legitimate promises for the future."

Was Claude Bernard correct in this opinion as to the "empty hands?" If scientific evidence is worth anything, it points to the appalling conclusion that, *notwithstanding all the researches of physiology, the chief forms of chronic disease exhibit to-day in England a greater fatality than thirty years ago*. In the following table I have indicated the average annual mortality, per million inhabitants, of certain diseases, *first*, for the period of five years from 1850 to 1854, and *secondly*, for the period twenty-five years later, from 1875 to 1879. The authority is beyond question; the facts are collected from the report to Parliament of the Registrar-general of England:

Average Annual Rate of Mortality in England, from Causes of Death, per One Million Inhabitants.

NAME OF DISEASE.	During Five Years, 1850-54.	During Five Years, 1875-79.
Gout,	12	25
Aneurism,	16	32
Diabetes,	23	41

Insanity,	29	57
Syphilis,	37	86
Epilepsy,	105	119
Bright's disease,	32	182
Kidney disease,	94	114
Brain disease,	192	281
Liver disease,	215	291
Heart disease,	651	1,335
Cancer,	302	492
Paralysis,	440	501
Apoplexy,	454	552
Tubercular diseases and diseases of the Respiratory Organs,	6,424	6,886
Mortality from above diseases:	**9,026**	**10,994**

This is certainly a most startling exhibit, when we remember that from only these few causes about half of *all* the deaths in England annually occur, and that from them result the deaths of two-thirds of the persons, of both sexes, who reach the age of twenty years.[25] What are the effects here discernible of Bernard's experiments upon diabetes? of Brown-Sequard's upon epilepsy and paralysis? of Flint's and Pavy's on diseases of the liver? of Ferrier's researches upon the functions of the brain? Let us appeal from the heated enthusiasm of the experimenter to the stern facts of the statistician. Why, so far from having obtained the least mastery over those malignant forces which seem forever to elude and baffle our art, they are actually gaining upon us; every one of these forms of disease is more fatal to-day in England than thirty years ago; during 1879 over sixty thousand *more* deaths resulted from these maladies alone than would have occurred had the rate of mortality from them been simply that which prevailed during the benighted period of 1850 to 1854! True, during later years there has been a diminished mortality in England, but it is from the lesser prevalence of zymotic diseases, which no one to-day pretends to cure; while the organic diseases show a constant tendency to increase. Part of this may be due to more accurate diagnosis and clearer definition of mortality causes: but this will not explain a phenomenon which is too evident to be overlooked.

"It is a fact," says the Registrar-general, in his report for 1879, "that while mortality in early life has been very notably diminished, *the mortality of persons in middle or advanced life has been steadily rising for a long period of years*." It is probable that the same story would be told by the records of France, Germany, and other European countries; it is useless, of course, to refer to America, since in regard to statistical information we still lag behind every country which pretends to be civilized.[26] Undoubtedly it would be a false assumption which from these facts should deduce retrogression in medical art or deny advance and improvement; but they certainly indicate that the boasted superiority of modern medicine over the skill of our fathers, due to physiological researches, is not sustained by the only impartial authority to which science can appeal for evidence of results.

What then is the substance of the whole matter? It seems to me the following conclusions are justified by the facts presented.

I. All experiments upon living animals may be divided into two general classes; 1st those which produce pain,—slight, brief, severe or atrociously acute and prolonged; and 2nd, those experiments which are performed under complete anæsthesia from which either death ensues during unconsciousness, or entire recovery may follow.

II. The majority of vivisections requisite for purposes of teaching physiological facts *may* be so carried on as to take life with less pain or inconvenience to the animal than is absolutely necessary in order to furnish meat for our tables. Those who would make it a penal offense to submit to a class of college students the unconscious and painless demonstration of functional activity of the heart, for example, and yet demand for the gratification of appetite the daily slaughter of oxen and sheep without anæsthetics, and without any attempt to minimize the agony of terror, fear and pain—may not be inconsistent. But it is a view the writer cannot share.

III. Prohibition of all experiments may be fairly demanded by those who believe that the enthusiastic ardor of the scientific experimenter or lecturer, will outweigh all considerations of good faith, provided success or failure of his experiment depend on the consciousness of pain. In other words, that the experimenter himself, as a rule, *cannot be trusted to obey the law, should the law restrict*.

This also is an extreme position.

IV. Absolute liberty in the matter of painful experiments has produced admitted abuses by physiologists of Germany, France and Italy. In America it has led to the repetition before classes of students of Magendie's extreme cruelties,—demonstrations which have been condemned by every leading English physiologist.

V. In view of the dangerous impulses not unfrequently awakened by the sight of pain intentionally inflicted, experiments of this kind should be by legal enactment absolutely forbidden before classes of students, especially in our Public Schools.

VI. It is not in accord with scientific accuracy to contend for unlimited freedom of painful experimentation, on the ground of its vast utility to humanity in the discovery of new methods for the cure of disease. On the contrary, so far as can be discovered by a careful study of English mortality statistics, physiological experiments upon living animals for fifty years back have in no single instance lessened the fatality of any disease below its average of thirty-five years ago.

VII. Vivisection, involving the infliction of pain is, even in its best possible aspect, a necessary evil, and ought at once to be restricted within the narrowest limits, and placed under the supervision of the State.

APPENDIX.

I.

For reasons sufficiently stated in the preceding pages, the writer does not advocate the total abolition of all experimentation. It is only fair to acknowledge, however, that very strong and weighty arguments in favor of legal repression have been advanced both in this country and abroad, some of which are herewith presented, as the other side of the question.

The cause of abolition has no more earnest and eloquent advocate than Miss Frances Power Cobbe of England. Through innumerable controversies with scientific men in the public journals, magazines and reviews, she has presented in awful array, the abuses of unlimited and uncontrolled experimentation on the continent of Europe, and the arguments in favor of total repression. The following letters, extracts from her public correspondence, will indicate her position.

TENDER VIVISECTION.

(To the Editor of the "Scotsman.")

1, Victoria Street, London, S. W.,
January 10, 1881.

SIR.—An Italian pamphlet, *Dell'Azione del Dolore sulla Respirazione* (The Action of Pain on Respiration), has just reached my hands, and as it is, I think, quite unknown in this country, I will beg you to grant me space for a few extracts from its pages. The pamphlet is by the eminent physiologist, Mantegazza, and was published by Chiusi, of Milan. Having explained the object of his investigations to be the effects of pain on the respiratory organs, the Professor describes (p. 20) the methods he devised for the production of such pain. He found the best to consist in "planting nails, sharp and numerous, through the feet of the animal in such a manner as to render the creature almost motionless, because in every movement it would have felt its torment more acutely" (*piantando chiodi acuti e numerosi* *attraverso le piante dei piedi in modo da rendere immobile o quasi l'animale, perché ad ogni movimento avrebbe sentito molto piu acuto il suo tormento*). Further on he mentions that, to produce still more intense pain (*dolore intenso*) he was obliged to employ lesions, followed by inflammation. An ingenious machine, constructed by "our" Tecnomasio, of Milan, enabled him likewise to grip any part of an animal with pincers with iron teeth, and to crush, or tear, or lift up the victim, "so as to produce pain in every possible way." A drawing of this instrument is appended. The first series of his experiments, Signor Mantegazza informs us, were tried on twelve animals, chiefly rabbits and guinea pigs, of which several were pregnant. One poor little creature, "far advanced in pregnancy," was made to endure *dolori atrocissimi*, so that it was impossible to make any observations in consequence of its convulsions.

In the second series of experiments twenty-eight animals were sacrificed, some of them taken from nursing their young, exposed to torture for an hour or two, then allowed to rest an hour, and usually replaced in the machine to be crushed or torn by the Professor for periods of from two to six hours more. In the table wherein these experiments are summed up, the terms *molto dolore* and *crudeli dolori* are delicately distinguished, the latter being apparently reserved for the cases when the victims were, as the Professor expresses it, *lardellati di chiodi*—("larded with nails").

In conclusion, the author informs us (p. 25) that these experiments were all conducted "*con molto amore e pazienza!*"—with much zeal and patience.

I am, etc.,
FRANCES POWER COBBE.

In a controversy with Dr. Pye-Smith, who had read a paper before the British Association, Miss Cobbe writes as follows to one of the public journals:

"Dr. Pye-Smith is reported to have said: 'Happily, the neccessary experiments were comparatively few.' Few! What are a "few" experiments? Professor Schiff in ten years experimented on 14,000 dogs, given over to him by the Municipality of Florence, and returned their carcases so mangled that the man who had contracted for their skins found them useless. He also experimented on pigeons, cats, and rabbits to the number, it is calculated, of 70,000 creatures; and he now asks for ten dogs a week in Geneva. All over Germany and France there are laboratories "using" (as the horrible phrase is) numberless animals, inasmuch as I have just received a letter stating that dogs are actually becoming scarce in Lyons, and it is proposed to breed them for the purpose of Vivisection. Be this true or not, I invite any of your readers to visit the office of the Victoria Street Society, and examine the volumes of splendid plates of vivisecting instruments, which will there be shown them, and then judge for themselves whether it be for a few experiments that those elaborate and costly inventions have become a regular branch of manufacture. Let them examine the volume of the English handbook of the physiological laboratory, the volume of Cyon's magnificent atlas, with its 54 plates, the *Archives de Physiologie*, with its 191 plates, the *Physiologische Methodik*, or Claude Bernard's *Leçons sur la Chaleur Animale*, with its pictures of the stoves wherein he baked dogs and rabbits alive; and after these sights of disgust and horror they will know how to understand the word "few" in the vocabulary of a physiologist. I am glad to hear that a German opponent of Vivisection recently entering a shop devoted to the sale of these tools of torture, was greeted by the proprietor with a volley of abuse: 'It is you and your friends,' he said, 'who are destroying my trade. I used to sell a hundred of Czermak's tables and other instruments for one I sell now.'

"Dr. Pye-Smith said: 'Many of the experiments inflicted no pain or injury whatever, and the great majority of the rest were rendered painless by the use of those beneficial agents which abolished pain and had themselves been discovered by experiments upon living animals.' As to the use of anæsthetics in annulling the agonies of mutilated animals, the audience ought to have asked Dr. Pye-Smith to explain whether he intended to refer to chloroform, or the narcotic morphia, or, lastly, to the drug *curare*. If he referred to chloroform, Dr. Hoggan tells from his own experience (*Anæsthetics*, p. 1), that 'nothing can be more uncertain than its influence on the lower animals; many of them die before they become insensible. Complete and conscientious anæsthesia is seldom even attempted, the animal getting at most a slight whiff of chloroform *by way of satisfying the conscience of the operator*, or enabling him to make statements of a humane character.' Even if it were conscientiously administered at the beginning of an experiment, how little would chloroform diminish the misery of Rutherford's dogs or Brunton's ninety cats, whose long-drawn agonies extended over many days? How little could it affect in any way the cases of starving, poisoning, baking, stewing to death, or burning,—like the twenty-five dogs over which Professor Wertheim poured turpentine and then set them on fire, leaving them afterwards slowly to perish? If Dr. Pye-Smith was thinking of morphia, the reader may refer to Claude Bernard's *Leçons de Physiologie Operatoire*, where he will find that great physiologists recommends its use; but at the same time mentions (as of no particular consequence) that the animal subjected to its influence still 'suffers pain.' I can hardly suppose, lastly, that Dr. Pye-Smith was secretly thinking of *curare*, and that he is one of those whom Tennyson says would

> "Mangle the living dog which loved him and fawned at his knee,
> Drenched with the hellish oorali."

It is bad enough to "mangle" a loving and intelligent creature without adding to its agonies the paralysis of the powers of motion, and the increased sensibility to pain occasioned by this horrible drug, which nevertheless Bernard, in the work above quoted, says is in such common use among physiologists, that when an experiment is not otherwise described, it may always be "taken for granted it has been performed on a curarized dog."

Finally, Dr. Pye-Smith says, "It was remarkable that the small residue of experiments in which some amount of pain was necessary were chiefly those in which the direct and immediate benefit to mankind was more obvious. He referred to the trying of drugs on animals, to discovering antidotes to poisons," etc. The bribe here offered to human selfishness is an ingenious one. "Let us," the physiologists say, "retain the right to put animals to torture, for it is very 'remarkable' that when we do so it is always in your interest!" Unluckily for this appeal to the meaner feelings of human nature, which these modern instructors of our young men are not ashamed to put forward, it is difficult for them to hit on any one instance wherein out of their "few" (million) experiments any good to mankind has been, even apparently, achieved. As Claude Bernard honestly said, at least as regards any benefit for suffering humanity, "*Nos mains sont vides*." As to the trying of drugs on animals, Dr. Pritchard, who is, I believe, the best living authority on the subject, told the Royal Commission (Minutes, 908), "I do not think that the use of drugs on animals can be taken as a guide to the doses or to the action of the same drugs on the human subjects." As to the discovery of antidotes to poison, the only man who seems on the verge of any success is the brave and noble fellow who has been trying such experiments not on animals but on himself.

In conclusion, I must add one word on Dr. Pye-Smith's last sentence, namely, "that legislation against vivisection is injurious to the best interests of the community." Sir, I know not what vivisectors deem to be the best interests of the community. For my part I do not reckon them to be the influence of drugs, nor yet susceptible of being carved out with surgical instruments. I do not think that they consist in escape from physical pain, nor even in the prolongation for a few years of our little earthly life. I hold that the best interests of the community are the moral and immortal interests of every soul in such community, namely, the conquest of selfishness, cowardice, and cruelty, and the development of the god-like sense of justice and love—the growth of the divinest thing in human nature, the faculty of sympathizing with the joys and sorrows of all God's creatures. Believing these to be "the best interests of the community," I ask, without hesitation, for the suppression of this abominable trade, which can best be described as "Pitilessness practised as a profession." If vivisection be indeed the true method of studying physiology, if physiology cannot be advanced except by vivisection, if chemical observation and microscopic research be useless for the purpose, and nothing but the torture of animals and the demoralization of men will suffice for its progress—then, in God's name, I say, let physiology stop at the point it has reached, even till the day of doom.—I am, Sir, with apologies for the length of this letter, yours, etc.

Frances Power Cobbe

Certainly, as regards the ethics of vivisection, nothing more eloquent has ever been written than this closing paragraph.

In a letter to the London TIMES in December, 1884, Miss Cobbe writes as follows:

TO THE EDITOR.

SIR,—In your article on this subject on Saturday last you called upon the opponents of vivisection to answer certain questions. As I have been intrusted for many years with the hon. secretaryship of the leading anti-vivisectionist society, I beg to offer you the following replies to those questions:—

You ask first, Do we "deny that vivisection is capable of yielding knowledge of service to man?" We are not so rash as to deny that any practice, even the most immoral conceivable, might possibly yield knowledge of service to man; and, in particular, we do not deny that the vivisection of human beings by the surgeons of classic times, and again by the great anatomists of Italy in the 15th century, may very possibly have yielded knowledge to man, and be capable, if revived, of yielding still more. We have, however, for a long time back called on the advocates of the vivisection of dogs, monkeys, &c., to furnish evidence of the beneficial results of their work, not as setting at rest the question of its morality, but as an indispensable preliminary to justify them in coming into the court of public opinion as defendants of a practice obviously (as the Royal Commissioners reported) "liable from its very nature to great abuse."

We must be excused if we now hold it to be demonstrated that, whether vivisection be or be not "capable of yielding useful knowledge," it certainly yields only a scanty crop of it. Were there anything like an abundant harvest, such a sample as this would not have been produced with so much pomp for public scrutiny. In short, we think with Dr. Leffingwell that, "if pain could be measured by money, there is no mining company in the world which would sanction prospecting in such barren regions."

You ask us, Sir, secondly, "Do we affirm that the benefit of mankind is not an adequate or sufficient justification for the infliction of pain on animals?" We have two answers to this question.

Assuming that by vivisection benefits might be obtained for human bodies, we hold that the evil results of the practice on human minds would more than counterbalance any such benefits. The cowardice and pitilessness involved in tying down a dog on a table and slowly mangling its brain, its eyes, its entrails; the sin committed against love and fidelity themselves when a creature capable of dying of grief on his master's grave is dealt with as a mere parcel of material tissues, "valuable for purposes of research"—these are basenesses for which no physical advantages would compensate, and the prevalence of such a heart-hardening process among our young men would, we are convinced, detract more from the moral interests of our nation than a thousand cases of recovery from disease would serve those of a lower kind. Even life itself ought not to be saved by such methods, any more than by the cannibalism of the men of the "Mignonette."

Our second answer is yet more brief. We do not "deny that the benefit of man is a sufficient justification for inflicting pain upon animals," provided that pain is kept within moderate bounds, nor yet to taking life from them in a quick and careful manner. But we do deny the right of man to inflict torture upon brutes, and thus convert their lives from a blessing into a curse. Such torture has been inflicted upon tens of thousands of animals by vivisection; and no legislation that ingenuity can devise will, we believe, suffice to guard against the repetition of it so long as it is sanctioned in any way as a method of research. The use of vivisection—if it have any use—is practically inseparable from abuse. We therefore call upon our countrymen to forego the poor bribes of possible use which are offered to them, and of which we have now seen a "unique and impressive" example, and generously and manfully to say of vivisection as they once said of slavery "We will have none of it."

I am, Sir, yours, etc.,
FRANCES POWER COBBE.

Hengwrt, Dolgelly, Dec. 28, 1884.

II.

[*Report of American Anti-vivisection Society, Jan. 1888.*]

"There remain two grounds to adopt: one the total abolition of all experiments; the other the total abolition of all *painful* experiments. This latter position, which is the one that Dr. Bigelow of Boston and Dr. Leffingwell have assumed, has engaged our attention for a long time; but, after bestowing upon it careful consideration, we feel that we must give it up as impracticable. To secure immunity from pain there must be absolutely perfect anæsthesia. This can be only obtained in two ways: one is by trusting to the experimenter himself to give sufficient of the anæsthetic; the other to insist that an assistant shall be present for the express purpose of keeping the animal under perfect anæsthesia. Now is it anyway likely that either of these conditions would be observed?"

III.
[*From the "Therapeutic Gazette," Detroit, Aug., 1880.*]

"Vivisection is grossly abused in the United States. * * We would add our condemnation of the ruthless barbarity which is every winter perpetrated in the Medical Schools of this country. History records some frightful atrocities perpetrated in the name of Religion; but it has remained for the enlightenment and humaneness of this century to stultify themselves by tolerating the abuses of the average physiological laboratory—all conducted in the name of Science. There is only one way to progress in Therapeutics; and that is by clinical observation; the noting of the action of individual drugs under particular diseased conditions. He who has the largest practice and is the keenest observer, and the most systematic recorder of what he sees, does the most to advance Medicine."

IV.
[*From editorial in "The Spectator," London, July 17, 1880.*]

"A memorial for the absolute abolition of vivisection has been presented to Mr. Gladstone with a great many most influential signatures attached. For our own part, were the experiments on the inoculation of animal diseases excepted,—experiments which, we venture to say, have sometimes proved of the greatest value to animals themselves,—we should, on the whole, be content to go with the abolitionists, not because we think all experiments, especially when conducted under strict anæsthetics, wrong, but because when they are permitted at all it is so extremely difficult to enforce properly and fully humane conditions. Dr. A. Leffingwell has sufficiently shown in the able paper in the July *Scribner's Magazine*, how extremely few remedies of value have resulted from this awfully costly expenditure of anguish. 'If pain could be estimated in money' he justly says, 'no corporation would be satisfied with such a waste of capital.' Take, as the single illustration of this most weighty sentence, Dr. Leffingwell's statement that what the late Dr. Sharpey called 'Magendie's infamous experiment' on the stomach of the dog, has been repeated 200 times without establishing to the satisfaction of scientific physiologists the theory for which that act of wickedness was first committed. No wonder the society for the Protection of Animals from Vivisection goes to extremes."

FOOTNOTES

[1] Report of American Anti-Vivisection Society, Jan'y 30, 1888.

[2] See Appendix, page 83.

[3] "Report of the Royal Commission on the Practice of Subjecting Live Animals to Experiments for Scientific Purposes." Question No, 175. Reference to this volume will hereafter be made in this article by inserting in brackets, immediately after the authority quoted, the number of the question in this report from which the extract is made.

[4] "Human Physiology," by John Elliotson, M. D., F. R. S. (page 448).

[5] "Medical Times and Gazette," October 5, 1872.

[6] A Text-book of Human Physiology, designed for the use of Practitioners and Students of Medicine, by Austin Flint, Jr., M. D. D. Appleton & Co. New York: 1876 (page 722).

[7] Page 738.

[8] Page 585.

[9] Page 710.

[10] Page 403.

[11] Pages 269-70.

[12] Page 282.

[13] Page 489.

[14] Page 629.

[15] Page 463.

[16] Pages 470-71.

[17] Flint: "Text Book on Human Physiology" (page 641).

[18] Dalton's "Human Physiology" (page 466).

[19] Flint (pages 639-40).

[20] The contradictory opinions ascribed to most of the authorities quoted in this article are taken directly from the "Report of the Royal Commission on the Practice of Subjecting Live Animals to Experiments for Scientific Purposes,"—a Blue-Book Parliamentary Report.

[21] "He feels the pain, but has lost, so to speak, the idea of self defense." Leçons de Physiologie opératoire, 1879, p. 115.

[22] Text-Book of Human Physiology, p. 595.

[23] "A Text-Book of Human Physiology." By Austin Flint, Jr. M. D. New York, 1876. Page 589; see also page 674.

[24] See Hansard's Parliamentary Debates, June 20, 1876.

[25] In 1879 the total mortality in England, above the age of twenty, from *all causes* whatsoever, was 287,093. Of these deaths, the number occasioned by the sixteen causes above named, was 191,706, or almost exactly two-thirds.

[26] Even Japan, a country we are apt to consider as somewhat benighted, has far better statistical information at hand than the United States of America.

TRANSCRIBER'S NOTES

1. Footnotes have been moved from the middle of a paragraph to the end of this e-text.

2. Some obvious punctuation errors in the text have been silently corrected, for example, missing period at a paragraph end, etc.

3. The following misprints have been corrected:
    "sufering" corrected to "suffering" (page 14)
    "anæthetics" corrected to "anæsthetics" (page 48)

4. Other than the corrections listed above, printer's inconsistencies in spelling, punctuation and hyphenation have been retained.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”

- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.

- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain "Defects," such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the "Right of Replacement or Refund" described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations ($1 to $5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a